**Gustavo J. Borjas**

## EDUCATION

2023-2031 Case Western Reserve University School of Medicine MSTP Program

- Doctor of Medicine (expected 2031)
- Doctor of Philosophy (expected 2029)

2017-2021 Emory University College of Arts and Sciences
Bachelor of Science

- Major in Biophysics
- Major in Biology
- Summa Cum Laude
- Thesis Title: "Transcriptional Roadblocking by the Bacteriophage 186 DNA Wrapping Protein"

## HONORS AND AWARDS

2021 Emory Honors College, Honors Thesis, *summa cum laude*
2020 Sigma Pi Sigma Physics Honors Society
2019 Initiative for Maximizing Student Development NIH Fellowship

## RESEARCH & WORK EXPERIENCE

2023 – Case Western Reserve University
PI: John Letterio
Project: "Novel Immunomodulatory Strategies Targeting Myeloid Cells in Cancer and Autoimmune diseases"

2022 – 2023 University of Chicago, Pritzker School of Molecular Engineering
PI: Jeffrey Hubbell
Project: "Novel RNA-editing Enzyme for Cancer Immunotherapy"
Project: "Novel Engineered Cytokine to Precisely Target Atherosclerotic Plaques"

- Immunotherapy development under Jeffrey Hubbell and Lisa Volpatti focused on 1.) delivering RNA editing enzymes for cancer immunotherapy and 2.) engineering precision cytokines for atherosclerosis
- Performed Immunological assays (ELISA, ELISpot, Flow Cytometry, IHC, IF)
- Protein engineering of antibody cytokine conjugates: design, production, and characterization
- Worked with mouse models of cancer and atherosclerosis
- Microscopy and image analysis of fluorescent and brightfield images

2021 – 2022 H. Lee Moffitt Cancer Center & Research Institute, Genitourinary Oncology/Immunology

PI: Roger Li/Jose Conejo-Garcia
Project: "Immune Monitoring of a Phase Ib Clinical Trial in Muscle Invasive Bladder Cancer based on a Novel Oncolytic Virus"

- Immune monitoring of clinical trials under the direction of Drs. Roger Li and Jose Conejo-Garcia
- Cancer Immunology research focused on understanding the changes in the bladder tumor microenvironment in response to oncolytic virus and immune checkpoint blockade
- Analysis of multiplex immunofluorescence images, genomic sequencing, and immunohistochemistry
- Performed immunological assays with primary human tumor samples (ELISA, ELISpot)
- Protocol writing for Human Biospecimen-use in compliance with IRB and IBC guidelines

2017 – 2021 Emory University, Department of Physics
PI: Laura Finzi
Project: "Transcriptional Roadblocking by the Bacteriophage 186 DNA Wrapping Protein"

- Biophysicla research involving DNA transcriptional regulation where a wrapping protein similar to histones was studied using single-molecule techniques
- Emphasis on Atomic Force Spectroscopy techniques and image analysis with MATLAB

Summer 2019 University of South Florida, Department of Physics
PI: Martin Muschol
Project: "Amyloid Formation Kinetics Revealed by Atomic Force Microscopy"

- Biophysical research involving amyloid formation kinetics
- Emphasis on Atomic Force Microscopy

## BIBLIOGRAPHY

Peer Reviewed Articles


1. Volpatti, L. R., Norton de Matos, S., Borjas, G., Beckman, T. N., Reda, J. W., Watkins, E. A., ... & Hubbell, J. A. (**2026**). LDL-binding IL-10 reduces vascular inflammation in atherosclerotic mice. *Nature Biomedical Engineering*, 1-12.

2. Li, R., Villa, N. Y., Yu, X., Johnson, J. O., Borjas, G., Dhillon, J., ... & Conejo-Garcia, J. R. (**2025**). Oncolytic immunotherapy with nivolumab in muscle-invasive bladder cancer: a phase 1b trial. *Nature Medicine*, 31(1), 176-188.

3. Kang, S., Mansurov, A., Kurtanich, T., Chun, H. R., Slezak, A. J., Volpatti, L. R., Chang, K., Wang, T., Alpar, A. T., Refvik, K. C., Hansen, I. O., Borjas, G.J., ... & Hubbell, J. A. (**2024**). Engineered GM-CSF polarizes protumorigenic tumor-associated macrophages to an antitumorigenic phenotype and potently synergizes with IL-12 immunotherapy. *Journal for Immunotherapy of Cancer*

4. Kang, S., Mansurov, A., Kurtanich, T., Chun, HR., Slezak, A., Volpatti, L., Chang, K., Wang, T., Alpar, A., Refvik, K., Borjas, G., Shim, H., Hultgren, K., Gomes, S., Solanki, A., Ishihara, J., Swartz, M., Hubbell, J. (**2024**). Engineered IL-7 synergizes with IL-12 immunotherapy to prevent T cell exhaustion and promote memory without exacerbating toxicity. *Science Advances*

5. Harro, CM., Sprenger, KB., Powers, JJ., Innamarato, PP., Anadon, CM., Zhang, Y., Chario, RA., Biswas, S., Manda, G., Mine, JA., Cortina, C., Nagy, MZ., Martin, AL., Handley, KF., Borjas, GJ., Chen, P., Pinilla-Ibarz, J., Sokol, L., Yu, X., Conejo-Garcia, JR. (**2023**) Sezary Syndrome Originates from Heavily Mutated Hematopoietic Progenitors. *Blood Advances*

6. Li, R., Sexton, W., Dhillon, J., Berglund, A., Naidu, S., Borjas, G., Rose, K., Kim, Y., Wang, X., Conejo-Garcia, J., Jain, R., Poch, M., Spiess, P., Pow-Sang, J., Gilbert, S., Zhang, J. (**2023**). A Phase 2 Study of Durvalumab for Bacillus Calmette-Guerin (BCG) Refractory Urothelial Carcinoma In Situ of the Bladder. *Clinical Cancer Research*

7. Lu, Y., Voros, Z., Borjas, G., Henderson, C., Shearwin, K., Dunlap, D., Finzi, L. (**2022**). RNA polymerase efficiently transcribes DNA-scaffolded, cooperative bacteriophage repressor complexes. *FEBS Letters.*

8. Borjas, G. (**2021**). Transcriptional Roadblocking by the Bacteriophage 186 DNA Wrapping Protein. *Emory Theses and Dissertations*.

9. Hasecke, F., Niyangoda, C., Borjas, G., Pan, J., Matthews, G., Muschol, M., & Hoyer, W. **(2021)**. Protofibril-Fibril Interactions Inhibit Amyloid Fibril Assembly by Obstructing Secondary Nucleation. *Angewandte Chemie International Edition, 60*(6) 3016.

10. Barton, J., Arias, D. S., Niyangoda, C., Borjas, G., Le, N., Mohamed, S., & Muschol, M. **(2019)**. Kinetic Transition in Amyloid Assembly as a Screening Assay for Oligomer-Selective Dyes. *Biomolecules*, *9*(10), 539.

Poster Presentations

Borjas, G., (**2026**). CDK5 Silencing in a Neural-Crest-Derived Tumor. TRCCC

Borjas, G., (**2022**). Oncolytic virus + Nivo induces a TLS driven immune response in Bladder Cancer. Moffitt Cancer Center Research Symposium

Borjas, G., (**2021**). Transcriptional Roadblocking by the Wrapping Protein. Poster presented and recognized at Emory's initiative for maximizing student development annual 5 minute presentation talks.

Borjas, G., Muschol, M., Barton, J. **(2019)**. Characterizing Amyloid Morphology and Interactions between Alternate Forms. Poster presented at a NIH sponsored conference at the end of a Research Experiences for Undergraduates (REU) program.

## MEMBERSHIPS

Phi Delta Epsilon International Co-ed Medical Fraternity
Sigma Pi Sigma Physics Honors Society

**Languages:** English and Spanish native speaker